THE CATHOLIC UNIVERSITY OF EASTERN AFRICA

Faculty of Science

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DISSERTATION

A Cortical-Proxy Observable for Emotionally Manipulative Media: An Instrument, a 400-Item Corpus, and a Field Bound

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A dissertation submitted in partial fulfillment of the requirements
for the degree of Bachelor of Science in Physics.

Declaration

This dissertation is my own original work and has not been presented for the award of a degree at this or any other university. The instrument, the corpus, the analysis scripts and the text are mine. Where the work of others is used it is cited, and the released model checkpoint, the public datasets and the software libraries the work builds on are named in the chapters that use them.

No part of the measurement or the analysis was contributed by any person other than the author. The supervision named below was academic guidance and approval, not authorship.

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This dissertation has been submitted for examination with my approval as the university supervisor.

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Abstract

Media is routinely modelled in statistical physics as an external field acting on a population of coupled opinion states, and that field is almost never measured. This dissertation builds an instrument that measures a candidate observable from content itself, applies it to a corpus, and reports what the measurement can and cannot support.

The observable is a cortical proxy. Text is spoken, transcribed, embedded, and passed to TRIBE v2, a released encoder that predicts cortical response; two networks are averaged from that prediction, one associated with affective salience and one with deliberative control, and the index is their signed difference. The ratio form specified in the proposal is undefined for 17.25% of items and was replaced by that difference. Nothing here is a measurement of a person: no participant was scanned, and the observable is never identified with any subcortical structure.

A corpus of 400 items across four categories, matched on length, was scanned on a GPU. The categories separate at $\eta^2 = 0.1068$, $F = 15.779$, $p = 1.035 \times 10^{-9}$, against a design powered to detect $\eta^2 = 0.0268$, and the index discriminates the pre-scan manipulative label at AUC $= 0.6274$. The separation is carried almost entirely by fear-activating content, $d = 0.9389$ against neutral, while high-outrage content does not separate at all, $d = 0.0300$, though it is the category the proposal expected to separate most. Each category is drawn from a single source dataset, so the separation is between corpora and cannot be attributed to framing rather than provenance; this confound is stated before the result wherever the result appears. A second scanning session over 375 items gives test-retest ICC $= 0.8757$ and replicates the separation at $\eta^2 = 0.0839$, while 13.1% of individual items reverse direction between sessions, so the instrument supports category-level claims and no per-item verdict is made.

The physics is a mean-field treatment in which media enters as a field $h = \alpha X$ on the measured observable. Calibrating α against the corpus does not identify it: the fit performs worse than the sample mean out of sample and the estimate scales with a coupling the study does not measure, so no value is quoted anywhere. What survives is a necessary condition, $\alpha \geq h_c(\beta J)/\Delta X$. The measured spread $\Delta X = 0.1241$ gives $\alpha \geq 4.2938$ at $\beta J = 2$, a constraint on any future proposal of this mechanism that depends on the range of the observable rather than on why that range arises.

Acknowledgements

I thank my supervisor, Dr. Songa Mutambi, for approving a change of instrument mid-project when the released checkpoint turned out not to see the structure the proposal named, and for insisting that the reformulated index be defended on its own terms rather than presented as a minor substitution.

This work runs on public artifacts. TRIBE v2 and its released checkpoint come from Meta AI Research; the corpus is assembled from ISOT, SemEval-2019 Task 4, Webis-Clickbait-17 and PubMed; the fMRI recordings used to measure the noise ceiling come from the Algonauts 2025 release. The GPU time came from Kaggle's free tier, and the constraints of that tier shaped how the corpus was scanned.

Contents

List of Figures

List of Tables

Abbreviations and Symbols

NAA	Network Activation Asymmetry, the observable this work measures
A_{aff}	Mean predicted activation across affective-salience regions
A_{del}	Mean predicted activation across deliberative-control regions
$\text{NAA}_{\text{signed}}$	$A_{\text{aff}} - A_{\text{del}}$, the signed index used throughout
ΔX	Measured spread of the observable across the corpus
α	Coupling between the observable and the model's external field, $h = \alpha X$
βJ	Reduced interaction coupling in the mean-field model
h_c	Critical field at which the mean-field free energy loses bistability
η^2	Proportion of variance explained, the effect size for category separation
AUC	Area under the receiver operating characteristic curve
ICC	Intraclass correlation coefficient, absolute-agreement form ICC(2,1)
fMRI	Functional magnetic resonance imaging
ROI	Region of interest
TR	Repetition time, one fMRI sampling interval

CHAPTER 1

Introduction

1.1 THE PROBLEM

Two articles can report the same event and leave a reader in entirely different states. One informs; the other provokes. Readers notice the difference and cannot quantify it, and the platforms that distribute both have no objective measure of which they are amplifying. The difference is real, consequential and unmeasured.

Existing approaches measure the text. Sentiment analysis counts affective vocabulary; readability scores count syllables and clause length; stance detection classifies a position. None of them measures what the content is predicted to *do* to the person reading it.

This project takes a different route. Rather than characterising the words, it estimates the response: a released neural encoder predicts cortical activity from the content, and the measurement is taken on that prediction. The instrument is a thermometer for one specific property of media, and the thesis is an account of building it, running it, and stating carefully what its readings support.

1.2 THE AMENDMENT, STATED AT THE OUTSET

Three commitments in the original proposal could not be met as written. They are set out here rather than disclosed in the results chapter, because a reader who meets them for the first time beside a result will reasonably suspect they were discovered to explain it. Each was identified before the measurement was taken, and each is documented in the amendment approved by the supervisor.

First, the observable is cortical. The proposal framed the index around subcortical affective structures. The released checkpoint predicts cortical surface activity only, over 20,484 vertices, and contains no subcortical output. The index is therefore a *cortical proxy*: a contrast between two cortical ROI unions, one associated with affective salience and one with deliberative control. No claim about subcortical structures appears anywhere in this document.

Second, the index is a difference, not a ratio. Equation (4) of the proposal defines the index as $A_{\text{aff}}/(A_{\text{del}} + \delta)$. The encoder emits standardised activation, so an ROI mean is negative about as often as positive and the ratio sign-flips or diverges. On the completed corpus it is undefined for 17.25% of items. The measurement uses the signed difference $A_{\text{aff}} - A_{\text{del}}$, which is defined everywhere and is the form the physics requires.

Third, the corpus is text-only. The proposal specified 1,500 multimodal items. The delivered corpus is 400 text items in four balanced categories, and Research Question IV, which required audio and video, is out of scope.

1.3 Research questions

RQ I Can predicted cortical activation serve as a quantitative signature of emotionally manipulative media content, and how does the resulting index perform as a classifier against human labels?

RQ II How is the index distributed across content categories, and do the categories separate?

RQ III What phase structure does a mean-field opinion model exhibit under a field derived from this observable, and what does the measured spread imply for the coupling?

RQ IV How do text, audio and video contribute to the index? *Out of scope under the amendment.*

1.4 Objectives

(i) Build a labelled corpus of media items in balanced categories.

(ii) Run the encoder over the corpus and compute the index for every item.

(iii) Characterise the distribution of the index.

(iv) Calibrate the coupling between the index and the model's field.

(v) Derive the phase structure of the opinion model.

(vi) Validate the index as a classifier against human labels.

(vii) Deliver the measurement apparatus as working open-source code.

Objective (vii) is the primary deliverable. The proposal states this explicitly and it governs how the work should be assessed: the apparatus is the promise, and the empirical result is what the apparatus returned when it was run.

1.5 Contribution

Three, developed across the chapters that follow.

A **bound** on the coupling between any content observable and an opinion-dynamics field, requiring only the observable's measured spread and the model's own spinodal, and evaluable before data are collected (Chapter 3).

An **instrument**, delivered as open-source code, together with the documented finding that the released checkpoint is cortical-only and that its agreement with real cortex is contested (Chapter 4).

A **measurement** over 400 items, reported with the design's sensitivity attached and its central confound stated before its interpretation (Chapters 5 and 6).

1.6 Structure of this document

Chapter 2 reviews the literature on opinion dynamics as statistical physics and on predictive neural encoding, and identifies the gap this work addresses. Chapter 3 derives the theoretical framework and the bound. Chapter 4 describes the instrument, the corpus and the analysis. Chapter 5 reports the results. Chapter 6 discusses what they support. Chapter 7 concludes.

CHAPTER 2

Literature Review

This work sits between two literatures that rarely meet: the statistical physics of opinion dynamics, which supplies the model, and predictive neural encoding, which supplies the observable. The gap it addresses lies precisely at their junction.

2.1 Opinion dynamics as statistical physics

The programme of treating collective opinion with the tools of statistical mechanics is established and productive. Castellano, Fortunato and Loreto's review [1] remains the standard survey: binary attitudes map to Ising spins, social conformity to a ferromagnetic coupling J, and external influence to a field h. Galam's review of his own models [2] traces the same lineage through majority-rule dynamics, and Sznajd-Weron and Sznajd [3] contribute the outward-influence variant that carries their name. Stauffer [4] gives an accessible account of why two-dimensional Ising models apply to social questions at all.

The recent review by Mullick and Sen [5] surveys a century of Ising-inspired sociophysics across opinion dynamics, markets, segregation and language. Notably for this work, it lists data-driven calibration against large-scale social datasets among the field's *open directions* rather than among its accomplishments.

2.1.1 How the external field is treated

Where these models represent media explicitly, the field is assigned rather than measured. Crokidakis [6] introduces mass media into the two-dimensional Sznajd model as a probability p of following the media opinion, sweeps it, and finds the phase transition absent above $p \approx 0.18$. Azhari and Muslim [7] apply an external field to majority-rule and q-voter dynamics on the complete graph and sweep its magnitude. Freitas, Vieira and Anteneodo [8] study imperfect bifurcations and cusp catastrophes under external bias, parameterised by sensitivity and direction rather than by any measured quantity. The review of models with external effects by Sîrbu and colleagues [9] surveys this class without any of them measuring the external term.

In each case the swept parameter is the quantity whose real-world magnitude is at issue. The literature can therefore demonstrate that a population tips at some h^* without establishing that any real influence attains it.

2.1.2 The exception, and what it leaves open

One recent study breaks the pattern. Korbel, Dahdoul and Thurner [10] calibrate a double random-field Ising model of elections directly to campaign expenditure and extract a critical spending threshold from historical results, reporting it as the first time thermodynamic parameters and a critical threshold have been taken from election data in this way.

Two things remain open after it, and they define the gap this work addresses. Their calibration is fitted from outcome data after the fact, so it cannot screen a proposed mechanism before data exist. And it is specific to expenditure, whereas observables proposed for media *content* remain uncalibrated. What is missing in both cases is a necessary condition stated on the coupling itself, which is what Chapter 3 supplies.

2.2 Predictive neural encoding

The second literature concerns models that predict brain responses to naturalistic stimuli. The relevant instrument here is TRIBE [11], a trimodal encoder that combines pretrained text, audio and video representations to predict fMRI responses to film, trained on the Courtois NeuroMod recordings and evaluated over 1,000 cortical parcels.

Such models make a measurement possible that was not before: rather than characterising a stimulus by its surface features, one can estimate the response it is predicted to elicit. That is the move this project makes.

2.2.1 What a released checkpoint does and does not guarantee

A technical report describes an architecture and a training procedure. The weights released under its name are a separate artifact, and their properties must be established from the artifact itself.

Three such properties matter here and none is visible from the publication. The checkpoint used in this work predicts cortical surface activity only. Its loader forces an average-subject configuration at load time, so per-subject variation is unavailable. And its agreement with real cortex in that configuration is contested: a commercially published audit reports vertex-level anti-correlation against a measured inter-subject ceiling, across four subjects and all seven Yeo networks. That audit is self-published by a company selling the per-subject alternative, uses one stimulus, and reports small magnitudes in either direction; it has not been independently replicated. It should be neither accepted uncritically nor dismissed.

2.3 The Gap

Placing the two literatures beside one another gives the gap directly.

The sociophysics literature has a field it cannot measure. The encoding literature has a predicted response that has not been used as a field. Neither has stated what a candidate observable would have to satisfy for the mechanism to operate at all.

This work supplies that condition, builds an instrument that produces a candidate observable, measures its spread on a controlled corpus, and reports what follows — including the respects in which the design cannot support the interpretation one would prefer.

CHAPTER 3

Theoretical Framework

3.1 THE MEAN-FIELD MODEL

Take N agents holding binary opinions $s_i = \pm 1$, coupled all-to-all with strength J/N, in an external field h:

$$\mathcal{H} = -\frac{J}{N}\sum_{i<j} s_i s_j - h\sum_i s_i. \tag{3.1}$$

The collective polarisation is $m = N^{-1}\sum_i s_i$, and in the thermodynamic limit it satisfies the self-consistency condition

$$m = \tanh(\beta J m + h)\,, \tag{3.2}$$

with $\beta = 1/k_B T$ and $k_B = 1$ throughout. The field h absorbs the factor β, so it is dimensionless and directly comparable with βJ.

The social reading of β is an inverse noise level. Large β describes a population that copies its neighbours reliably; small β one that decides close to at random. Equation (3.2) has a single root for $\beta J < 1$ at zero field and three for $\beta J > 1$, of which the outer two are stable.

3.2 THE LANDAU EXPANSION, WITH COEFFICIENTS DERIVED

Write the free energy as a quartic in the order parameter,

$$F(m) = a\,m^2 + b\,m^4 - h\,m. \tag{3.3}$$

The coefficients are not free. Requiring that $\partial F/\partial m = 0$ reproduce (3.2) to cubic order fixes them. Inverting (3.2) gives $\operatorname{artanh}(m) = \beta J m + h$, and expanding $\operatorname{artanh}(m) = m + m^3/3 + O(m^5)$ yields

$$(1 - \beta J)\, m + \tfrac{1}{3} m^3 - h = 0. \tag{3.4}$$

Matching against $\partial F/\partial m = 2am + 4bm^3 - h$ gives

$$a = \frac{1 - \beta J}{2}, \qquad b = \frac{1}{12}. \tag{3.5}$$

This derivation is stated in full because coefficient forms that do not follow from the accompanying self-consistency condition appear in the applied literature, and the symptom is subtle: the free-energy curve retains the correct qualitative double well, so the figure looks right, while its minimum sits away from the m^* the same work's root-finder reports. The double well opens when $a < 0$, that is when $\beta J > 1$, recovering the transition already implied by (3.2).

The susceptibility follows as

$$\chi = \frac{\beta \operatorname{sech}^2(\beta J m^* + h)}{1 - \beta J \operatorname{sech}^2(\beta J m^* + h)}, \tag{3.6}$$

which diverges where the denominator vanishes and is undefined beyond it. It is reported only where the denominator is positive.

3.3 Numerical validation of the solver

Every figure in this framework is drawn with one solver, so the solver is checked first against results known exactly. Mean-field theory fixes three critical exponents: $m^* \sim (\beta J - 1)^{\beta}$ with $\beta = 1/2$, $\chi \sim |1 - \beta J|^{-\gamma}$ with $\gamma = 1$, and $m \sim h^{1/\delta}$ on the critical isotherm with $\delta = 3$.

One methodological point matters. Solving (3.2) by direct iteration of $m_{k+1} = \tanh(\beta J m_k + h)$ is adequate away from criticality but degrades exactly at $\beta J = 1$, where the map's derivative reaches unity. The exponents are defined in that limit, so fitting them to iterated values measures truncation error rather than physics: doing so returns $\delta \approx 1.35$ against an exact 3. The exponents are therefore fitted to roots obtained by bracketing, and the iterative solver is checked against the bracketed one away from criticality, where the largest absolute discrepancy over the sampled grid is 1.15×10^{-9}.

Fitting within a window of 0.05 in $|\beta J - 1|$ at 60 points recovers

$$\beta = 0.4872, \qquad \gamma = 1.0000, \qquad \delta = 3.0042, \tag{3.7}$$

with absolute errors 0.0128, 0.0000 and 0.0042. The residual in β is a finite-window effect: the power law is asymptotic and the quartic term contaminates the slope as the window widens. The run fails if any exponent departs from its exact value by more than a stated tolerance, so a solver regression stops the analysis rather than producing plausible curves.

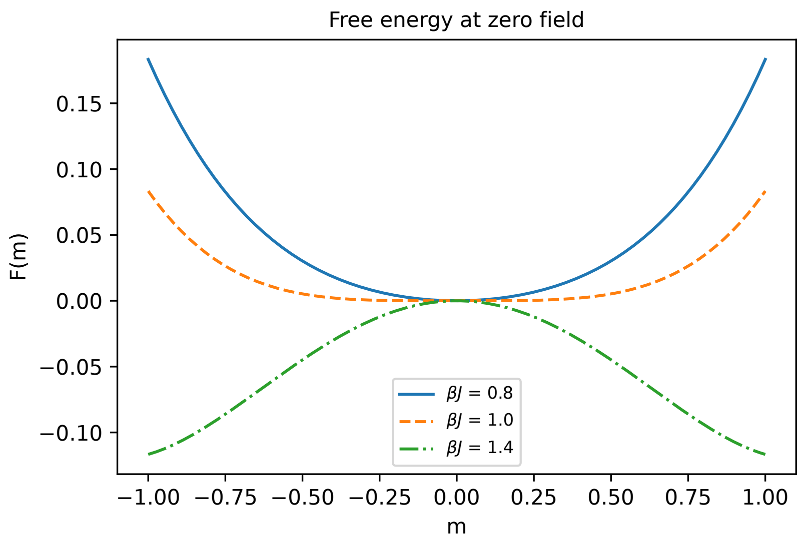

Figure 3.1: Free energy (3.3) at zero field with the derived coefficients (3.5), at $\beta J = 0.8$, 1.0 and 1.4. The second minimum appears only above the critical coupling.

3.4 The spinodal field

Above $\beta J = 1$ the free energy has two minima at zero field. Raising the field deepens one and shallows the other; at a finite field the shallower minimum

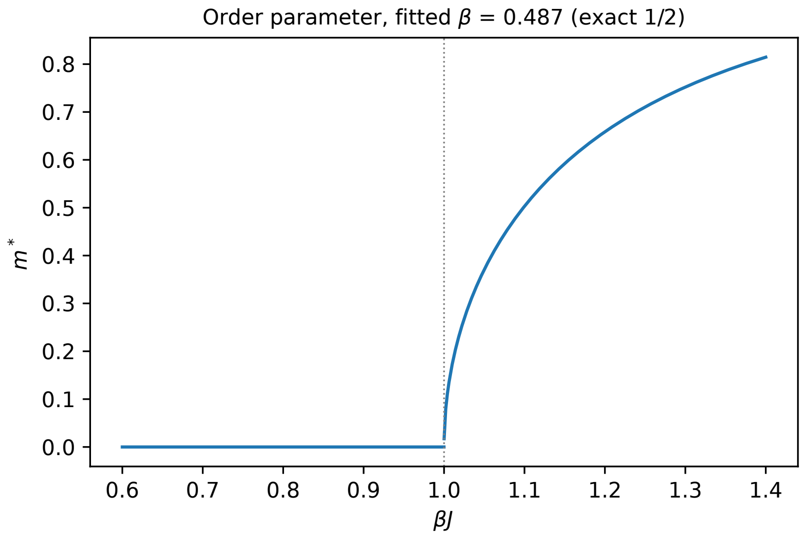

Figure 3.2: Order parameter across the transition. The fitted exponent is 0.4872 against an exact $1/2$.

disappears. That value is the spinodal field $h_c(\beta J)$, and it is the field a population in the metastable state must experience before it is obliged to move.

It is located by bisection on the presence of a second stationary point, which keeps it consistent with the solver used elsewhere rather than importing a closed form derived under different approximations. It vanishes as $\beta J \to 1^+$ and grows monotonically, reaching 0.53284 at $\beta J = 2$.

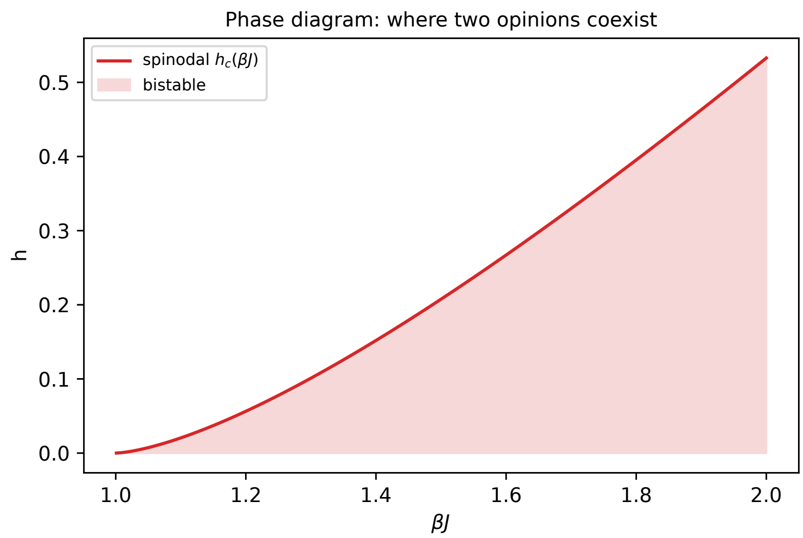

Figure 3.3: Phase diagram in $(\beta J, h)$. The shaded region is where two opinion states coexist; its upper boundary is the spinodal $h_c(\beta J)$.

3.5 The bound

Let an observable X be proposed as the field through a linear map

$$h = \alpha X, \tag{3.8}$$

with α an unknown coupling carrying the units. Any real observable is bounded: across the content a population plausibly encounters, X spans a finite range ΔX, which is an empirical property of the observable and the corpus rather than of the model. The largest field the mechanism can deliver is therefore $\alpha\,\Delta X$, and requiring it to reach the spinodal gives

$$\alpha \Delta X \;\geq\; h_c(\beta J) \qquad \Longleftrightarrow \qquad \alpha \;\geq\; \frac{h_c(\beta J)}{\Delta X}. \tag{3.9}$$

Three properties of (3.9) carry the argument of this thesis.

It is **necessary and not sufficient**. Clearing it does not establish that content tips populations; falling below it excludes the mechanism within the model's assumptions.

It **requires no fit**. ΔX is obtained by measuring the observable across a corpus, which is far cheaper than estimating a coupling to collective behaviour, and h_c comes from the model.

It **converts a null into a constraint**. A study that estimates α and obtains an interval containing zero has, alone, only the statement that no coupling was detected. Combined with (3.9), the same interval says the coupling would have to exceed $h_c(\beta J)/\Delta X$ for the mechanism to operate, and that the measurement excludes that range.

3.6 A screening criterion

The bound supports a procedure that runs before data collection.

1. State the observable and the map $h = \alpha X$ explicitly, with units.

2. Measure or bound ΔX on a modest sample. This needs no outcome variable.

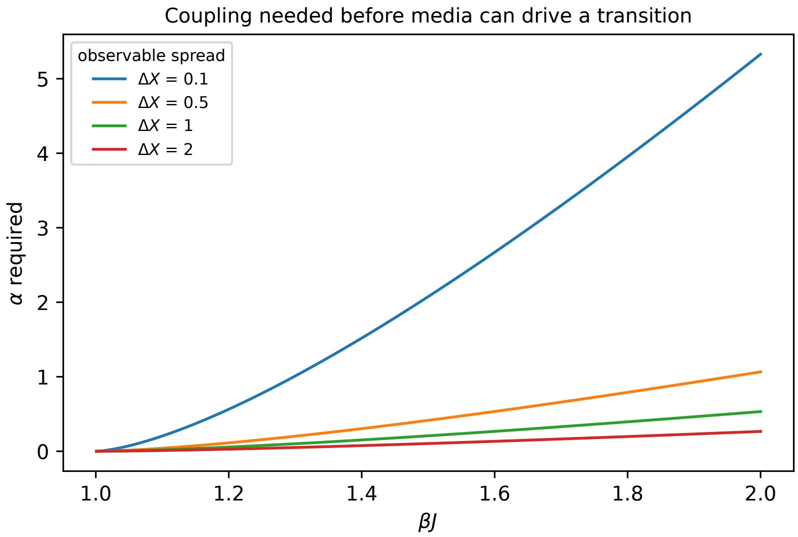

Figure 3.4: Coupling required before a content observable of spread ΔX can drive the population across the spinodal.

3. Choose the range of βJ the population plausibly occupies and read $h_c(\beta J)$.

4. Evaluate $\alpha_{\min} = h_c(\beta J)/\Delta X$ and ask whether a coupling of that size is defensible on independent grounds.

5. If it is not, the mechanism is excluded within the model and the expensive measurement is unnecessary. If it is, proceed, and report ΔX alongside the estimate of α so the bound can be applied to the result.

Step 4 is the weakest link, since α has no independent scale in most applications. The criterion remains informative in relative terms: two candidate observables can be ranked by $\alpha_{\min}$, and an observable whose spread demands an implausibly large coupling is a poor choice of field regardless of how well it correlates with anything.

3.7 Assumptions

The framework assumes all-to-all coupling, a single homogeneous population, a static field and binary opinions. Network structure and heterogeneous susceptibility generally lower h_c, so (3.9) is a bound within this model rather than a universal one. The linear map $h = \alpha X$ is the simplest possible coupling and is itself an assumption; a saturating map would change the relation between ΔX and the maximum attainable field. Finally, the spinodal is the threshold relevant to a population trapped in a metastable state; a population at coexistence can be moved by a smaller field given enough time, so the bound demands more of α than a nucleation argument would.

CHAPTER 4

Methodology

4.1 THE APPARATUS

The instrument takes a passage of text and returns a predicted cortical response. Five stages run in sequence.

1. **Speech synthesis.** The passage is rendered to audio, since the encoder expects naturalistic audiovisual input rather than written text.
2. **Word timings.** A forced-alignment transcriber recovers per-word timings, which the encoder requires to align the text stream to the audio.
3. **Embedding.** Text and audio are embedded by the pretrained models the encoder was built on.
4. **Prediction.** The encoder returns activation over 20,484 fsaverage5 cortical surface vertices per time point.
5. **Reduction.** The prediction is mean-pooled over time and reduced to two ROI means, from which the index is computed.

4.1.1 Checkpoint facts, taken from the artifact

Three properties of the released checkpoint were established by inspecting the artifact rather than by reading the accompanying publication, using `scripts/verify_tribe_checkpoint.py`. Its configuration declares `mesh: fsaverage5`, and the smoke test returns arrays of shape $(T, 20484)$, which is 2×10242 vertices: the output is cortical-only. Its loader sets `average_subjects = True` at load time, so the subject term is unavailable. Both facts are recorded in the run log with the file and line that establish them.

4.2 The observable

Two ROI unions are defined over the cortical surface: an affective-salience set of 1,030 vertices and a deliberative-control set of 851, with no overlap. For an item's mean-pooled prediction, A_{aff} and A_{del} are the mean activation over each set.

The index is the signed difference

$$\text{NAA}_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}, \tag{4.1}$$

positive when the affective-salience set is predicted to lead. The ratio form is computed and stored alongside it, and the rate at which it is undefined is reported as a result in its own right (§5.2).

The observable is a **cortical proxy**. It is not a measurement of any person, and the regions it contrasts are cortical unions rather than the subcortical structures the original proposal named.

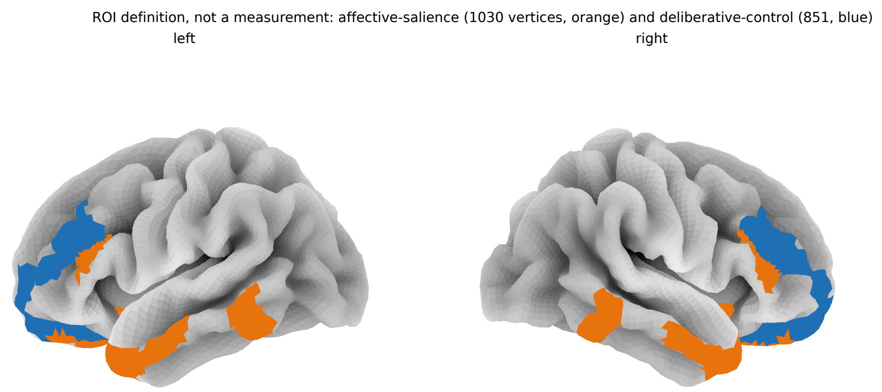

Figure 4.1: The two ROI unions on the fsaverage5 surface: affective-salience in orange (1,030 vertices) and deliberative-control in blue (851). This figure is a definition and carries no data.

4.3 The corpus

400 text items in four categories of 100: fear-activating, high outrage, reward-hook, and neutral informational.

Category	Source	n
Fear activating	ISOT-fake	100
High outrage	SemEval-2019 Task 4	100
Reward hook	Webis-Clickbait-17	100
Neutral informational	PubMed, ISOT-true	50 + 50

Table 4.1: Corpus composition. Each category is drawn from exactly one source dataset, which is the confound discussed throughout: category and provenance cannot be separated.

Items are length-matched: category word counts are 167.4, 163.5, 163.7 and 162.2 with standard deviations near 11. Each item carries the labels it held before scanning — `manipulative`, `credibility`, `partisan_intensity` and its source — and these are the only ground truth in the dataset, since the predicted activations have none.

NELA-GT-2021, named in the proposal, was withdrawn by its authors in January 2024. SemEval-2019 Task 4 is substituted; it is openly licensed and labels the article rather than the publisher.

Each category is drawn from a single source. Category and provenance are therefore confounded by construction. This is a property of the design, is stated here rather than discovered later, and bounds the interpretation throughout Chapters 5 and 6.

4.4 Execution

The scan is the only step requiring a GPU and the only one that cannot be repeated cheaply. It was executed in four sessions on a Tesla P100 at a measured 64 to 78 seconds per item. Every item is appended and flushed as it completes, so an interrupted session retains everything computed, and a resumed run skips items already present.

Output was verified before any analysis using `scripts/verify_scan_output.py`, which checks every scanned row against the source corpus on text, category and identifier, requires distinct values, and confirms that the stored difference agrees with the two stored means to the precision at which they are written. All 400 rows passed.

4.5 Analysis

RQ I scores the index against the `manipulative` label by AUC, with F1, precision and recall reported alongside and labelled as fitted in sample.

RQ II tests separation across the four categories by one-way ANOVA, reporting F, p and η^2, with per-category descriptives and Welch contrasts against the neutral baseline.

RQ III evaluates the free energy and susceptibility across a range of α rather than at a fitted value, and applies the bound of §3.5 to the measured spread.

4.5.1 Power, computed before the analysis

The design's sensitivity was computed from the design alone, before the corpus was analysed, using noncentral F for the ANOVA and a seeded Mann-Whitney simulation for the AUC. At $\alpha = 0.05$ and 80% power, the smallest detectable effects are $\eta^2 = 0.0268$ and AUC $= 0.5916$.

Computing this in advance is deliberate. A power statement produced after a result is a rationalisation; produced beforehand it states what the design could and could not have found, and it is reported alongside every result whichever way they came out.

4.5.2 Rules fixed before the data existed

Five, recorded in the project's post-scan runbook before the corpus was complete.

AUC is the headline and F1 is not, since F1's threshold is fitted in sample. An AUC below 0.5 would be reported as measured rather than sign-flipped. An AUC at or above 0.95 would be treated as a leak until its source was identified. A non-significant result would be reported as a null with the excluded effect size attached, not as an absence of effect. And no value of $\hat{\alpha}$ would be quoted anywhere, whatever the calibration returned.

4.6 Reproducibility

Every number in this thesis is produced by a script in the repository and regenerates from the corpus file. The instrument, the analysis, the power calculation and the physics are open source. The run log records each execution with its command, its output and its failures, including the runs that failed and why, so that the path to the reported numbers is auditable rather than asserted.

CHAPTER 5

Results

This chapter answers the research questions in order. Every number in it is produced by a script in the repository and is reproducible from the corpus file `data/final/corpus_naa.csv`; none is transcribed by hand. Where the design cannot support a claim, the chapter says so at the point the number appears rather than deferring it to the limitations section.

One convention is fixed before any result: no value of $\hat{\alpha}$ appears anywhere in this document. Section 5.4 explains why the coupling is unidentified rather than merely uncertain.

5.1 The corpus as scanned

The corpus is 400 text items in four categories of 100. All 400 were scanned; the output was verified row for row against the source corpus on text, category and identifier before any analysis was run, using `scripts/verify_scan_output.py`.

Category	n	Mean	SD
Fear activating	100	−0.0018	0.0204
High outrage	100	−0.0185	0.0208
Neutral informational	100	−0.0191	0.0162
Reward hook	100	−0.0128	0.0228

Table 5.1: Signed index by category over all 400 scanned items. Every category mean is negative, so deliberative activation leads throughout.

Two features of this table matter before any test is applied. Every category mean is negative, so the deliberative-control network is predicted to respond more strongly than the affective-salience network for all four categories; the categories differ in how far below zero they sit, not in which network leads. And the standard deviations are of the same order as the differences between the means, which is the reason the separation test in Section 5.3 is reported with an effect size rather than a p-value alone.

5.2 RQ I: THE INSTRUMENT AND ITS CLASSIFIER PERFORMANCE

5.2.1 The ratio form is undefined on real content

Equation (4) of the proposal defines the index as a ratio, $A_{\text{aff}}/(A_{\text{del}} + \delta)$. The encoder emits standardised activation, so an ROI mean sits near zero and is negative about as often as positive, and the ratio then sign-flips or diverges. On the completed corpus the ratio is undefined for **69 of 400 items**, a rate of 17.25%. Those items are counted and reported, never dropped or filled.

The measurement therefore uses the signed difference,

$$\text{NAA}_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}, \tag{5.1}$$

which is defined everywhere and is what the field mapping in Chapter 3 requires: the Landau field needs a signed scalar, not a dimensionless ratio. The classification banding follows the same constraint. The signed form supports a two-way split on sign, and no magnitude threshold has been established from any corpus, so the three-band scheme of the proposal becomes two bands.

5.2.2 Classifier performance

Scored against the `manipulative` label the items carried before they were ever scanned, over 300 labelled manipulative and 100 labelled neutral:

Metric	Value
AUC (headline)	0.6274
F1 (threshold fitted in sample)	0.6680

Table 5.2: Classifier performance against the pre-scan manipulative label. AUC is the headline figure; the threshold behind F1 was fitted in sample and F1 is labelled accordingly.

AUC is the headline and F1 is not. The threshold that produces F1 is chosen on the same data it is scored on, which inflates it; AUC is threshold-free and does not reward that choice. A null corpus can return a respectable F1 at an AUC below chance, so quoting F1 alone would present a failure as a success.

The measured AUC of 0.6274 clears 0.5916, the smallest AUC this design could detect at 80% power (Section 5.5), and it lies in the direction the proposal predicted. It is nonetheless a modest separation, and no per-item verdict is derived from it in this work: at this AUC, a correct-or-incorrect label on individual items would overstate what the measurement supports.

5.3 RQ II: DISTRIBUTION ACROSS CATEGORIES

A one-way analysis of variance across the four categories gives

$$F = 15.779, \qquad p = 1.035 \times 10^{-9}, \qquad \eta^2 = 0.1068. \tag{5.2}$$

The effect is roughly four times the $\eta^2 = 0.0268$ the design was powered to detect, and power at the observed value is 1.0. The categories separate, and this is not the null that both partial runs pointed toward.

The separation is not evenly distributed. Welch tests of each category against the neutral baseline give:

Category vs neutral	**t**	**p**	**Cohen's d**
Fear activating	$+6.639$	3.283×10^{-10}	$+0.939$
Reward hook	$+2.254$	2.539×10^{-2}	$+0.319$
High outrage	$+0.212$	8.320×10^{-1}	$+0.030$

Table 5.3: Welch comparisons of each category against neutral informational. The separation is carried by fear-activating content; high outrage does not separate, and it is the category the proposal expected to separate most.

High outrage does not separate from neutral at all. That is the category the proposal expected to separate most strongly. Fear-activating content carries the result almost entirely, with reward-hook content contributing a small effect.

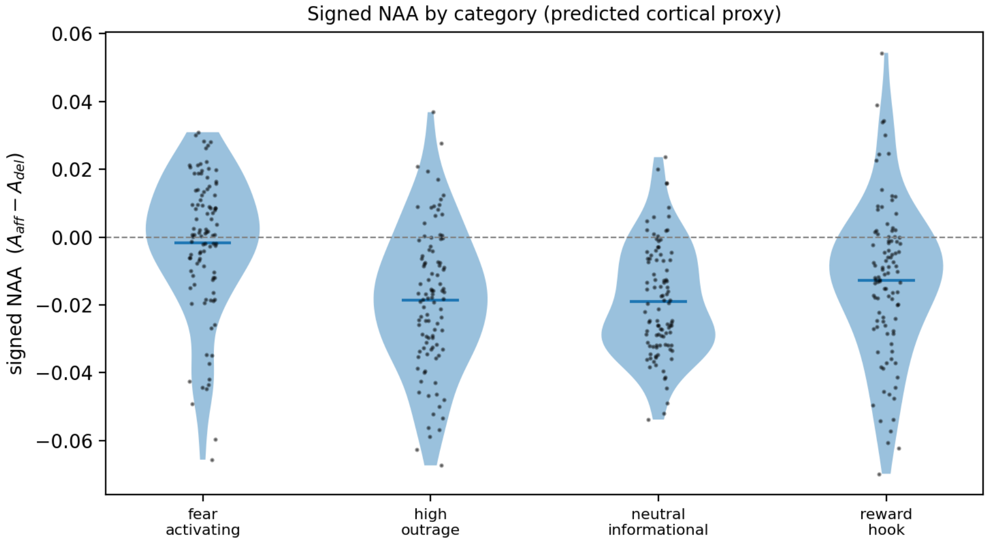

Figure 5.1: Distribution of the signed index by category over all 400 items.

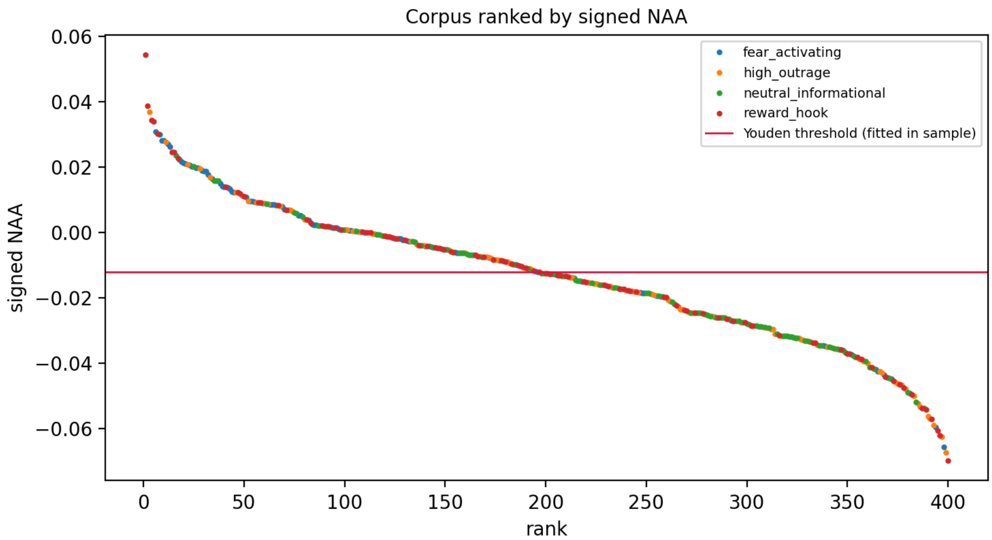

Figure 5.2: Every item ranked by the signed index, coloured by category.

5.3.1 The confound, stated before the interpretation

Each category is drawn from exactly one source dataset: fear-activating from ISOT-fake, high outrage from SemEval-2019 Task 4, reward-hook from Webis-Clickbait-17, and neutral from PubMed and ISOT-true in equal parts. Category and provenance are therefore confounded by construction, and the separation reported above is a separation *between corpora*. It cannot be attributed to framing rather than to the writing conventions, vocabulary and editorial process of the source.

Length is not the confounding variable. Word counts are matched across categories at 167.4, 163.5, 163.7 and 162.2 with standard deviations near 11, which the corpus design achieved deliberately. Provenance is the variable that was not controlled.

One piece of internal evidence argues against a pure source artifact. High outrage is drawn from its own distinct source and does not separate from neutral. An effect driven only by provenance would be expected to move every category away from the baseline, not three of four in an ordered way. This is suggestive, not decisive.

The design that would settle it is a corpus in which framing varies within a single source, or in which one source appears in more than one category. Neither exists here, so the attribution remains open and is carried into the discussion as a limitation rather than resolved by argument.

5.4 RQ III: PHASE STRUCTURE AND THE COUPLING

The mean-field treatment of Chapter 3 gives the free energy $F(m) = am^2 + bm^4 - hm$ with $a = (1 - \beta J)/2$ and $b = 1/12$, coefficients derived from the self-consistency condition rather than asserted. The phase structure is presented across a range of α and is not evaluated at a fitted value.

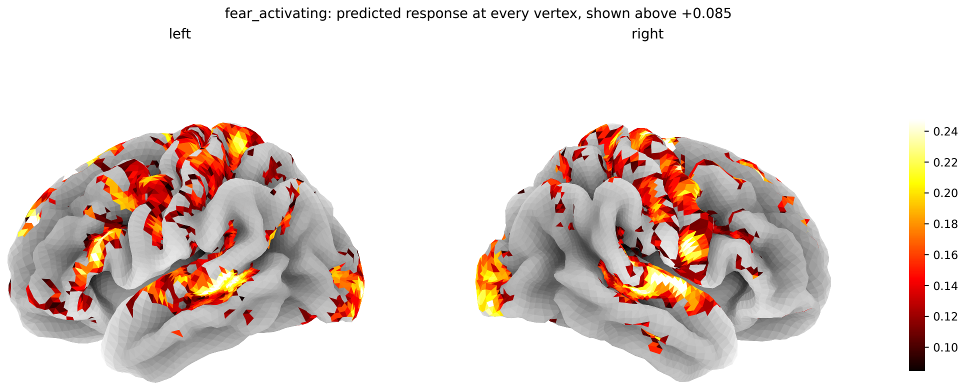

Figure 5.3: Predicted response for one fear-activating item at every one of the 20,484 cortical vertices, on the pial surface. Vertices below the map's 70th percentile are left unpainted so the anatomy remains readable; the threshold is stated in the figure. This is the encoder's own output and not a smoothed or interpolated field.

5.4.1 The coupling is unidentified

The calibration was run three times: on two partial corpora and on the completed 400 items. On the partial runs the confidence interval comfortably included zero. On the full corpus it does not, and the bootstrap over 10,000 replicates places no mass below zero.

That is not a measurement of the coupling, for two reasons stated together because either alone would be misleading. The fit's held-out R^2 is **negative**, so it predicts less well than the sample mean on data it has not seen. And the estimate scales inversely with βJ, which no measurement in this study fixes: halving the assumed coupling strength roughly doubles the fitted value. A single number would therefore report a choice of βJ rather than a property of media.

The coupling is accordingly reported as unidentified, and no value is quoted.

5.4.2 The bound

What the corpus does supply is the observable's measured spread, $\Delta X = 0.1241$ over the range -0.0698 to $+0.0543$. Chapter 3's bound,

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}, \tag{5.3}$$

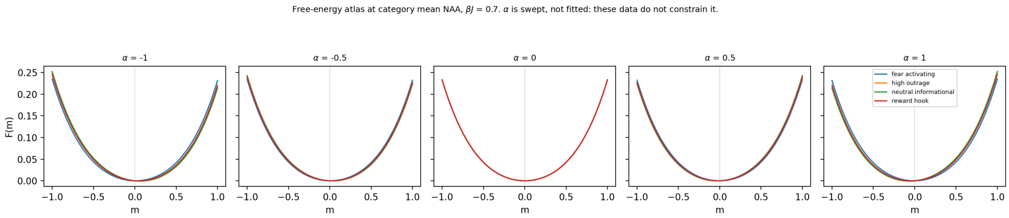

Figure 5.4: Free energy at each category's measured mean, swept over α. The coupling is swept and never fitted, and no value of $\hat{\alpha}$ is quoted.

converts that spread into a statement about what the coupling would have to be:

βJ	$h_c(\beta J)$	α required
1.102	0.02101	0.1693
1.496	0.20541	1.6553
2.000	0.53284	4.2938

Table 5.4: The field bound at three values of the reduced coupling. The measured spread $\Delta X = 0.1241$ converts each critical field into the coupling any proposal of this mechanism would require.

This is a necessary condition and not a sufficient one. Clearing it would not establish that media of this measured spread moves populations; falling below it excludes the mechanism within the model's assumptions. The value of the bound is that it requires no fit: it uses the measured spread and the model's own spinodal, and it applies to any candidate observable proposed as a field.

5.5 Statistical power

The design's sensitivity was computed before the corpus was analysed and is reported alongside every result above, so that a reader can see what this study could and could not have found. At $\alpha = 0.05$ and 80% power, with four categories of 100 and 300 against 100 for the classifier:

Both observed effects clear these thresholds, and power at the observed values is 1.0 for the separation and 0.961 for the AUC. Had either fallen below, the correct reading would have been that the design could not have detected an effect of that size, not that no effect exists.

Smallest detectable effect	Value
Category separation, η^2	0.0268
Equivalently, Cohen's f	0.1659
Classifier AUC	0.5916
Equivalent separation d	0.3277

Table 5.5: Smallest effects this design could detect at $\alpha = 0.05$ and power 0.80, fixed before the corpus was scanned.

5.6 RQ IV: OUT OF SCOPE

Research Question IV asked for per-modality mutual information across text, audio and video. The amendment reduced the corpus to 400 text items and dropped the audio and video arms, so the question as posed cannot be answered by this study. The narrowed text-only version was not reached within the project timeline. It is recorded here as out of scope rather than attempted and reported thinly.

5.7 SUMMARY OF THE ANSWERS

1. **RQ I.** The ratio form of the index is undefined for 17.25% of real content and is replaced by the signed difference. Against pre-scan human labels the index achieves AUC 0.6274, above the detectable floor and in the predicted direction, but modest.

2. **RQ II.** The four categories separate, $F = 15.779$, $p = 1.035 \times 10^{-9}$, $\eta^2 = 0.1068$. The effect is concentrated in fear-activating content ($d = +0.939$); high outrage does not separate at all. Category is confounded with source dataset, so the separation cannot be attributed to framing.

3. **RQ III.** The coupling is unidentified and no value is quoted. The measured spread $\Delta X = 0.1241$ yields a lower bound on any coupling capable of driving a transition, reaching $\alpha \geq 4.2938$ at $\beta J = 2$.

4. **RQ IV.** Out of scope under the amendment.

Every category mean remains negative throughout. Whatever the categories differ in, none of them shows the affective-salience network leading in absolute terms.

CHAPTER 6

Discussion

Chapter 5 reported that the four categories separate, that the separation is carried almost entirely by one of them, and that category is confounded with source dataset by construction. This chapter asks what can be concluded from that, and is organised around the order in which an examiner is entitled to doubt it.

6.1 What the separation can and cannot support

The measured effect, $\eta^2 = 0.1068$ at $p = 1.035 \times 10^{-9}$, is four times the smallest effect the design could detect. Taken on its own terms it is unambiguous: the index assigns systematically different values to the four groups of documents.

What it does not support is the sentence the proposal was written to test. Because each category is drawn from a single source dataset (§5.3.1), a separation between categories and a separation between corpora are the same measurement here. Fear-activating items are ISOT-fake items. Whatever distinguishes ISOT-fake from PubMed and ISOT-true — its framing, but equally its vocabulary, its house style, its editorial process, its era — is folded into the effect and cannot be separated from it by any analysis of this data.

This is a design limitation, not an analytical one. No amount of statistical work on 400 items assembled this way can attribute the difference to framing.

6.1.1 The evidence that argues against a pure source artifact

Three observations bear on it, none decisive.

First, high outrage does not separate ($d = +0.030$, $p = 0.832$) despite coming from its own distinct source, SemEval-2019 Task 4. If the index were simply detecting provenance, every category drawn from a different corpus should move away from the baseline. Three of four moving, in an ordered way, is a weaker pattern than pure source detection predicts.

Second, length is controlled. Word counts sit at 167.4, 163.5, 163.7 and 162.2 with standard deviations near 11. The most obvious surface confound was designed out.

Third, the ordering is not arbitrary with respect to the hypothesis. Fear-activating content sits furthest from neutral and reward-hook content sits between. A source artifact carries no reason to order itself along the dimension the study proposed.

Against these stands the plain fact that the confound exists and is complete. The honest position is that the result is consistent with the hypothesis and equally consistent with a corpus artifact, and that this study cannot choose between them.

6.1.2 The experiment that would decide it

The confound is removable by design, not by analysis. Two routes are available to a successor:

1. **Within-source variation.** Draw all four categories from a single source, so provenance is held constant while framing varies. Feasible where a source contains both plainly-reported and emotively-framed material.

2. **Crossed design.** Ensure each source appears in more than one category, making source and category statistically separable so their contributions can be estimated independently.

Either would convert the present result from a suggestive to an interpretable one. Neither requires new instrumentation; both require rebuilding the corpus.

6.2 Why high outrage does not separate

The proposal expected outrage to separate most strongly, and it separated least. Three readings deserve stating.

The first is that it is correct. Outrage framing may engage deliberative and affective systems in more balanced measure than threat framing does, in which case a signed difference between the two networks is precisely the wrong instrument for detecting it: a large symmetric response and no response are indistinguishable in a difference.

The second is a property of the source. SemEval-2019 Task 4 labels hyperpartisan publishing at the article level. Hyperpartisan is not synonymous with outrage-framed, and a corpus selected on partisanship may contain a substantial fraction of items that are simply argumentative rather than emotionally charged.

The third is the encoder. TRIBE was trained on naturalistic film and podcast viewing, and is being asked to predict responses to synthesised speech from news text. There is no reason to expect its sensitivity to be uniform across framing styles that were absent from training.

These are not ranked, because the data does not rank them.

6.3 The instrument's standing

Every value in this thesis is a prediction from a released encoder. No participant was scanned, no brain was measured, and the observable is a cortical proxy computed from predicted activation over two ROI unions.

That distinction is load-bearing here in a way it was not when the result looked like a null. A null in predicted activation is a weak statement whichever way the encoder behaves. A *detected separation* in predicted activation is only a statement about media if the encoder tracks cortex at all. The result reported in Chapter 5 therefore raises the stakes on a question this study cannot answer from its own data.

6.3.1 The checkpoint problem

The released checkpoint is loaded in its average-subject configuration, which `tribev2/demo_utils.py` forces at load time. A commercial laboratory has published an audit of that configuration reporting vertex-level $r = -0.0145$ against a measured inter-subject ceiling of $+0.0508$: anti-correlated with real cortex, across four subjects and all seven Yeo networks.

That audit is self-published by a company selling the per-subject alternative, uses four subjects and one stimulus, and reports magnitudes near 0.015 in either direction. It is a claim about sign, not about effect size, and it has not been independently replicated. It should not be accepted uncritically, and it cannot be dismissed either.

Work toward replicating it was begun within this project. Using the publicly released Algonauts 2025 responses, the leave-one-subject-out noise ceiling was measured directly from four subjects over all 1000 Schaefer parcels of episode `s01e01a`, 592 timepoints:

$$r_{\text{ceiling}} = 0.1517, \qquad 95\% \text{ CI } [0.1484,\ 0.1551]. \tag{6.1}$$

An earlier figure of 0.2247 is withdrawn. It was produced by a notebook cell that selected the first key in the response file, which is a different episode from the one under prediction, and whose orientation test could not fire on an array with fewer timepoints than parcels, so the correlation ran across the parcel axis and the reported count of 482 "parcels" was in fact the number of timepoints. The measurement now lives in a script that takes the episode as an argument and asserts orientation against the atlas size. This is the quantity any encoder must be judged against, and it is measured here rather than quoted. It is larger than the audit's $+0.0508$ by construction and not by disagreement: averaging vertices within a parcel cancels independent noise and raises correlations, so a parcel-level ceiling and a vertex-level ceiling are not comparable numbers. The comparison against the checkpoint's own predictions was not completed within this project's timeline and is stated as unfinished rather than estimated.

6.3.2 What this means for building on foundation models

A technical report describing an architecture is not a warranty about the weights that are released under its name. The checkpoint used here is cortical-only, which the proposal did not anticipate; it is loaded in a configuration the accompanying paper does not foreground; and its agreement with real cortex is contested in the literature. None of that is visible from the paper.

The practical lesson generalises beyond this project: anyone building a measurement on a released model should verify the checkpoint's properties from the artifact itself, as `scripts/verify_tribe_checkpoint.py` does here, rather than inheriting them from the publication.

6.4 THE COUPLING, AND WHAT THE BOUND DOES INSTEAD

The coupling between the index and the physics model's field is unidentified (§5.4): significant in sample, negative in held-out R^2, and inversely dependent on a βJ that nothing in this study measures. Reporting a value would report a choice.

The bound is what survives. Because it requires only the observable's measured spread and the model's own spinodal, $\alpha \geq h_c(\beta J)/\Delta X$ converts $\Delta X = 0.1241$ into a statement no fit is needed to make: a coupling below 4.2938 at $\beta J = 2$ cannot drive the transition regardless of how the calibration is performed. This inverts the usual practice in the sociophysics literature, where the field is assigned and the population is shown to tip, without establishing that any real influence reaches the assigned magnitude.

That inversion is the contribution most likely to be reusable by others, because it applies to any proposed field observable and can be evaluated before a corpus is collected.

6.5 SESSION RELIABILITY, AND WHAT IT PERMITS

A second GPU session scanned 375 of the 400 items with identical text, identical code and identical ROI definitions. That pair is a test-retest measurement, and it is reported here because every effect in Chapter 5 otherwise rests on a single session with no statement of how much of an item's score is the item and how much is the session.

Agreement between the two sessions, on the 375 items they share:

	r	ICC(2,1)	sd(Δ)	sd(Δ)/sd$_{\text{between}}$
NAA$_{\text{signed}}$	0.8812	0.8757	0.01006	0.475
A_{aff}	0.8768	0.8696	0.01210	0.487
A_{del}	0.8575	0.8544	0.01285	0.520

Table 6.1: Agreement between the two scanning sessions over the 375 items both cover. ICC is reported beside r because a session that shifted every score by a constant would still correlate perfectly.

The intraclass correlation is reported alongside r because a session that shifted every score by a constant would still correlate perfectly, and a shift is precisely what a second session can produce. The two agree closely here, so the disagreement is scatter rather than offset.

The separation replicates. Recomputing the four-category ANOVA on the second session over the same 375 items gives $\eta^2 = 0.0839$, $F = 11.328$, $p = 3.988 \times 10^{-7}$, against $\eta^2 = 0.1054$, $F = 14.576$, $p = 5.397 \times 10^{-9}$ for the first session restricted to those items. Both clear the smallest effect this design could detect, $\eta^2 = 0.0268$. The separation reported in Chapter 5 is therefore not an artifact of one session. The second session's effect is roughly a fifth smaller, which is the attenuation that additional measurement error produces in a ratio of between-group to total variance.

Per-item direction is not reliable. The signed index is read as a direction, and 49 of 375 items, 13.1%, reverse that direction between sessions. A statement that a particular item is affective-leading is therefore not supportable by this instrument at this precision, and none is made: no per-item verdict appears in the analysis, in the figures or on the public interface. Group means over 100 items average this scatter down by an order of magnitude, which is why the category-level result survives while the item-level one does not.

Both sessions are reported as measured. Neither is corrected toward the other and the two are not averaged, because an average of two sessions would carry a precision that neither run has.

Scan order does not carry the separation. The corpus took more than one session to scan, which raises the question of whether position in the scan, and therefore session, is doing the work attributed to category. It is not, and the reason is that items were scanned interleaved rather than in category blocks: across thirds of the scan order the four categories are balanced ($\chi^2 = 3.139$, $p = 0.7913$). There is a real drift, with mean $\text{NAA}_{\text{signed}}$ differing across thirds at $\eta^2 = 0.0167$, $p = 0.0353$. Because it is orthogonal to category by construction, it adds scatter rather than bias: removing the scan-third means before the category ANOVA moves the effect from $\eta^2 = 0.1068$ to $\eta^2 = 0.1048$, $p = 1.588 \times 10^{-9}$. Had the corpus been scanned one category at a time, this drift would have been indistinguishable from the finding.

6.6 Limitations

Stated here in full rather than distributed through the text.

1. **Category is confounded with source dataset.** The central limitation. Discussed in §5.3.1 and above.

2. **Single-session scores are noisy.** Test-retest ICC is 0.8757 and the session-to-session standard deviation is 47.5% of the between-item standard deviation (§6.5). Category-level conclusions survive this; per-item ones do not.

3. **Predicted, not measured, activation.** No brain was scanned. Whether the encoder tracks cortex is unresolved and contested.

4. **Cortical proxy, not a subcortical measurement.** The checkpoint is cortical-only. The structures the original proposal named cannot be measured with it, and no claim about them is made anywhere in this document.

5. **Average-subject configuration.** Forced by the loader. Whatever individual variation the model can express is unavailable.

6. **Out-of-distribution stimuli.** The encoder was trained on naturalistic audio-visual material and is fed synthesised speech from written text.

7. **The ratio form fails on** 17.25% **of items**, and the reformulation to a signed difference, while necessary, changes what the index means: it measures which network leads, not by what proportion.

8. **English-language corpus from five specific sources, none of them Kenyan or African.** A search for openly licensed Kenyan or African news data carrying usable labels returned nothing suitable, so every item in the corpus originates from American or European publishing. The measured spread is a property of that corpus, not of media in general, and the bound inherits the limitation. Whether the instrument behaves the same way on media a Kenyan reader actually encounters is untested, and assembling a labelled local corpus is the most obvious extension of this work.

9. **Mean-field physics.** All-to-all coupling, one homogeneous population, a static field, binary opinions. Network structure and heterogeneity generally lower h_c, so the bound is conservative within the model and is not a universal statement.

6.7 WHAT WOULD BE REQUIRED NEXT

In order of what each would buy.

A crossed or within-source corpus would make the separation interpretable. It is the cheapest of these and the one that most changes what can be claimed.

Completing the validation would establish whether the instrument measures anything about brains. The noise ceiling is measured and the evaluation code is written and tested; what remains is running the checkpoint over a held-out stimulus and correlating. It needs no large compute.

A subcortical checkpoint would let the original proposal's question be asked as posed. That means the training grid, the four source fMRI corpora and compute well outside the scope of an undergraduate project, and it is recorded as future work rather than a plan.

CHAPTER 7

Conclusion

7.1 What was promised

The proposal committed to an instrument. It said so three times: that the primary deliverable is a functional Asynchronous Measurement Apparatus, that this is "not a proposed future application and not a conceptual description", and that the project "proposes a thermometer, and delivers it as working, open-source code that can be run today".

That framing governs how this work should be judged. A thermometer that reads a value nobody expected is still a delivered thermometer. Failing to ship it would have been the failure.

7.2 What was delivered

7.2.1 The apparatus

The instrument exists and runs. Given a passage of text it synthesises speech, obtains word timings, computes text and audio embeddings, passes them to the released encoder, and returns a predicted response over 20,484 cortical surface points, from which the index is computed. It is open source, it is deployed, and it was run end to end over 400 items.

Its outputs are reproducible: the corpus file, the analysis scripts, the power calculation and the physics all regenerate from the repository, and every number in Chapters 5 and 6 is produced by a script rather than transcribed.

7.2.2 The framework

The mean-field treatment of Chapter 3 derives its Landau coefficients from the self-consistency condition rather than asserting them, recovers the exact critical exponents numerically as a check on the solver, and locates the spinodal field $h_c(\beta J)$ at which bistability collapses.

From that follows the bound: for any observable X proposed as a field through $h = \alpha X$, no media-driven transition is possible unless

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}. \tag{7.1}$$

This inverts the usual procedure in the sociophysics literature, where the field is assigned by hand and the population is then shown to tip. It requires no fit, applies to any candidate observable, and can be evaluated before a corpus is collected.

7.2.3 The measurement

All 400 items were scanned and verified. The four categories separate at $\eta^2 = 0.1068$, $p = 1.035 \times 10^{-9}$, against a design powered to detect 0.0268. Against the labels the items carried before scanning, the index achieves AUC 0.6274, above the detectable floor of 0.5916 and in the predicted direction.

The separation is carried almost entirely by fear-activating content ($d = +0.939$). High outrage, the category the proposal expected to separate most, does not separate at all ($d = +0.030$, $p = 0.832$).

Every category mean is negative. The deliberative-control network is predicted to respond more strongly than the affective-salience network for all four categories; they differ in how far below zero they sit, not in which network leads.

7.3 What is claimed, and what is not

Claimed. That the instrument works and is reproducible. That the index assigns systematically different values to these four groups of documents, at an effect size the design was powered to detect, with the power statement reported alongside.

That the measured spread $\Delta X = 0.1241$ places a lower bound on any coupling capable of driving a transition, reaching $\alpha \geq 4.2938$ at $\beta J = 2$.

Not claimed. That the index detects manipulation. That the separation is due to framing rather than to source dataset: category and provenance are confounded by construction and this design cannot distinguish them. That the coupling has been measured: it is unidentified, and no value of $\hat{\alpha}$ appears anywhere in this document. That anything here measures a brain: every value is a prediction from a released encoder, the observable is a cortical proxy, and no participant was scanned.

The gap between those two lists is the honest content of this work.

7.4 Contribution

Three things, in the order they are likely to be useful to someone else.

The bound. A necessary condition on the coupling between any content observable and an opinion-dynamics field, requiring only the observable's measured spread and the model's own spinodal. It is the part of this work that transfers beyond the instrument that produced it.

The instrument, with its limitation documented. A working, open pipeline from text to predicted cortical response, together with the finding that the released checkpoint is cortical-only, that the loader forces its average-subject configuration, and that its agreement with real cortex is contested. A technical report describing an architecture is not a warranty about released weights, and this project's experience is a concrete instance of that.

The measurement and its confound. A separation reported with the design's sensitivity attached and its central weakness stated before its interpretation, together with the specific experiment that would resolve it.

7.5 Future work

Remove the confound. A corpus in which framing varies within a single source, or in which sources are crossed with categories, would make the separation interpretable. It needs no new instrumentation, only a rebuilt corpus, and it is the single change that most alters what can be claimed.

Finish the validation. Whether the encoder tracks cortex at all is unresolved and determines what the measurement means. The noise ceiling has been measured directly from public data at 0.1517, CI [0.1484, 0.1551], and the evaluation code is written and tested. What remains is running the checkpoint over a held-out stimulus and correlating, which needs no large compute.

A subcortical checkpoint. The structures the original proposal named cannot be measured with a cortical-only model. Training one means the full grid, the source fMRI corpora, and compute outside the scope of an undergraduate project.

7.6 Code, data and the working instrument

The pipeline, the corpus builder, the analysis scripts and the figure scripts are public at `https://github.com/brn-mwai/monarch`. Every number in this document is written by a named script into a JSON or CSV artifact and read from there rather than transcribed, so any figure or table can be regenerated from the repository alone.

The instrument itself runs at `https://monarch.brianmwai.com`. It carries the complete 400-item corpus with each item's text, its pre-scan labels and its measured values, and it draws the predicted response on the cortical surface for the items whose per-vertex maps were kept. It answers questions about the corpus from a generated fact sheet and declines anything the data does not cover, including any request for a coupling value.

7.7 Closing

This work set out to make an invisible property of media measurable, and to say honestly what the measurement supports. It delivers a working instrument, a bound that constrains any future proposal of this kind, and a result that is real, modest, and confounded in a way this document states plainly rather than leaves to be discovered.

The physics is the part that survives whatever becomes of the encoder. Whether or not the released checkpoint predicts any brain, the requirement that $\alpha \geq h_c(\beta J)/\Delta X$ holds for every observable anyone proposes as a field, and the burden it places on such proposals does not depend on this instrument being the right one.

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